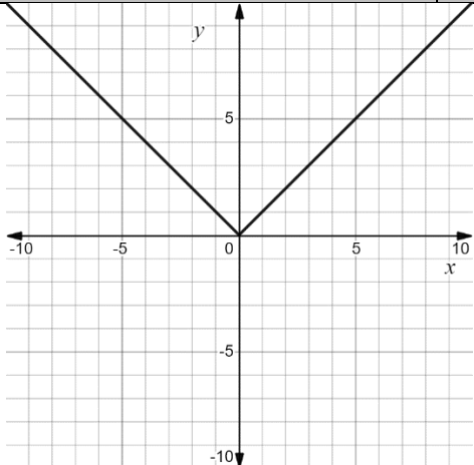


1. Select the function family for each.

Function		Function Types	
a.		☐ Absolute Value Function ☐ Cubic Function ☐ Linear Function ☐ Square Root Function	☐ Cube Root Function ☐ Exponential Function ☐ Quadratic Function
b.	$y = x^2$	☐ Absolute Value Function ☐ Cubic Function ☐ Linear Function ☐ Square Root Function	☐ Cube Root Function ☐ Exponential Function ☐ Quadratic Function

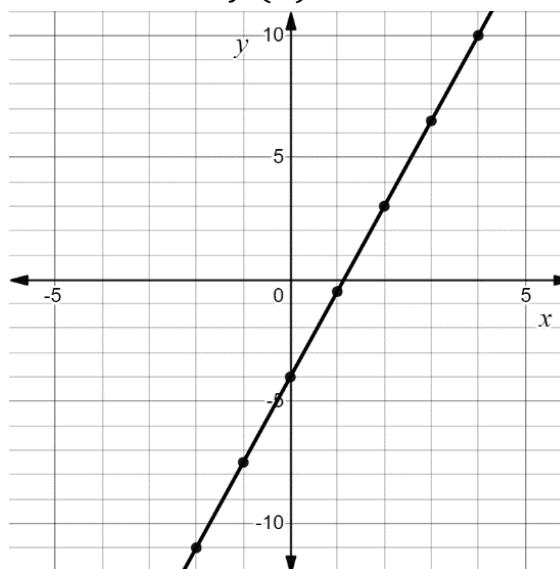
2. Find each value using the graph of the function $f(x)$.

a. $f(-2) =$ _____

b. $f(4) =$ _____

c. Find x when $f(x) = -7.5$.

$x =$ _____



3. Use the table of values representing $g(x)$ to find each value.

a. $g(-6) =$ _____

b. $g(4) =$ _____

c. Find x when $g(x) = 0$.

$x =$ _____

x	$g(x)$
-8	4
-6	1
-5	0
-3	-2
0	-4
2	-5
4	-6

4. The function $M(x) = 5x^2 - 160x + 6179$ models the number of miles, in millions, that U.S. passengers traveled on a train for the period 1970 to 2000, where x is the number of years since 1970.

- a. Find $M(10)$. Show your work for partial credit.

$M(10) =$	
-----------	--

- b. Interpret $M(10)$ for the given context.

5. The function $P(w) = 5,673w - 1,354$ models the relationship between a diamond's weight, w , in carats, and its price, P , in dollars.

Find $P(2)$. Show your work for partial credit.

$P(2) =$	
----------	--

6. Find the value of the variable that makes each equation true. For partial credit, show your work or explain your thinking.

Equation	Value of the Variable
a. $3^{\frac{21}{8}} \cdot 3^{\frac{11}{8}} = 3^x$	$x = \underline{\hspace{2cm}}$
b. $2^{24} = (2^y)^3$	$y = \underline{\hspace{2cm}}$
c. $61^{\frac{1}{3}} = \sqrt[k]{61}$	$k = \underline{\hspace{2cm}}$
d. $\frac{4^{\frac{3}{5}}}{4^{\frac{18}{5}}} = 4^n$	$n = \underline{\hspace{2cm}}$

7. Six expressions are shown.

A	B	C	D	E	F
$\sqrt{9^3}$	$(9^3)^{\frac{1}{2}}$	$\left(9^{\frac{1}{3}}\right)^2$	$(9)^{\frac{2}{3}}$	$(9)^{\frac{3}{2}}$	$(\sqrt{9})^3$

Select all of the expressions that are equivalent to 27.

- ☐ Expression A
- ☐ Expression B
- ☐ Expression C
- ☐ Expression D
- ☐ Expression E
- ☐ Expression F

8. Six expressions are shown.

A	B	C
$8\sqrt{2}$	$4\sqrt{8}$	$2 \cdot (16^{\frac{1}{2}})^3$
D	E	F
$(16)^{\frac{3}{2}} \cdot (2)^{\frac{1}{2}}$	$(4)^{\frac{3}{2}} \cdot (2)^{\frac{1}{2}}$	$2\sqrt{8}$

Select all of the expressions that are equivalent to $\sqrt{128}$.

- ☐ Expression A
- ☐ Expression B
- ☐ Expression C
- ☐ Expression D
- ☐ Expression E
- ☐ Expression F

9. Evaluate each expression. Show your work for partial credit.

a. $(36)^{\frac{1}{2}} \cdot (3^3)$ <table border="1" style="width: 100%;"><tr><td style="width: 20%; background-color: #cccccc; text-align: center; padding: 5px;">Value</td><td style="width: 80%; height: 40px;"></td></tr></table>	Value		b. $(6^{-1}) \left(9^{\frac{3}{2}}\right)$ <table border="1" style="width: 100%;"><tr><td style="width: 20%; background-color: #cccccc; text-align: center; padding: 5px;">Value</td><td style="width: 80%; height: 40px;"></td></tr></table>	Value	
Value					
Value					

10. Write a different equivalent expression for the expression.

$$\sqrt{75} + \sqrt{27} - \sqrt{48} = \underline{\hspace{2cm}}$$

11. Find the value of each expression. Show your work for partial credit.

a. $\sqrt{12} \cdot \sqrt{12}$ <table border="1" style="width: 100%;"><tr><td style="width: 20%; background-color: #cccccc; text-align: center; padding: 5px;">Value</td><td style="width: 80%; height: 40px;"></td></tr></table>	Value		b. $3\sqrt{72} + 7\sqrt{8}$ <table border="1" style="width: 100%;"><tr><td style="width: 20%; background-color: #cccccc; text-align: center; padding: 5px;">Value</td><td style="width: 80%; height: 40px;"></td></tr></table>	Value	
Value					
Value					